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Paper Title

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ABSTRACT Key points of intro (context and gap). Key point of our Methods. Key results. Key impact.

INDEX TERMS magnetic sensors, Hall effect

> **Examples**
>
> Here are two of my recent papers formatted with Quarto:
>
> - [1] is a tutorial paper on technical writing with Quarto.
> - [2] is a full technical paper, whose source is available on Gitlab.
>
> Note that there is now an official extension to format the output according to IEEE template

I. INTRODUCTION

A. CONTEXT

- Big-picture challenge in the world ("hook the audience").
- Zoom into a specific topic.
- Prior knowledge & work, scientific SoA
- Open research question & gap
  - there is a **need** to address this gap

B. OBJECTIVES / SENSOR RQTS

- High-level restrictions (IN vs OUT of scope)
- Specific measurable metrics

C. PREVIEW

- Preview the novel idea (concept, approach, ...) TIP: include a concept overview figure on page 1. See Figure 1.
- Highlight unique contributions.
- Preview paper structure: Section 1 describes ...

II. (RELATED WORKS)

In long paper, dedicated section to review the state of the art.

Figure 1: Key concept.

III. DESIGN

"Methods" if more appropriate. Use action verb to emphasize the procedure (the "recipe").

Figure 2 shows the architecture block diagram.

A. DESIGN OF ELEMENT 1

B. DESIGN OF ELEMENT 2

IV. IMPLEMENTATION

Pictures of realizations and/or experimental setups.

V. (EXPERIMENTAL) RESULTS

- Test 1
- Test 2

VI. DISCUSSION

- Key insights
- Comparison with previous work
- Highlight the strengths by explaining

Figure 2: Block diagram.

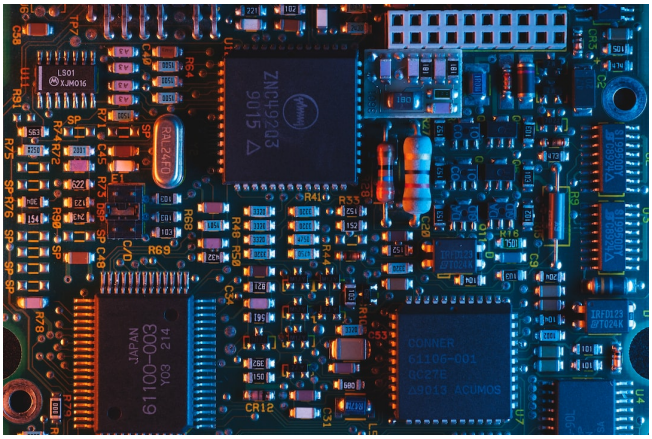

Figure 3: Picture of the prototype.

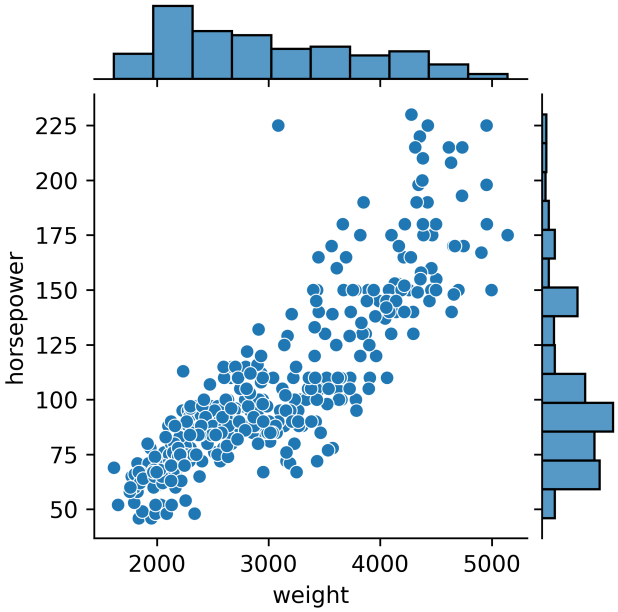

Figure 4: Result of test 1.

  - how the paper moves the SoA
  - how the paper addresses the original objectives
- Discuss limitations and recommendation for future work

VII. CONCLUSION

- Summary: methods & results & significance
- Extrapolate / speculate on the broad impact

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APPENDIX A: REFRESHER ON XXX

It might be needed to provide a refresher on some well-known background knowledge. Review of Theory X, Fundamentals of X, …

APPENDIX B: NOMENCLATURE / NOTATION

List of symbols. Sometimes included just after the intro.

REFERENCES

[1] G. Close, "Technical Writing and Publishing Data-Rich Articles with Quarto," Sep. 22, 2022. [Online]. Available: https://gael-close.github.io/2209-tech-writing.html

[2] B. Brajon, E. Gasparin, and G. Close, "A benchmark of integrated magnetometers and magnetic gradiometers," IEEE Access, vol. 11, pp. 115635–115643, 2023, doi: 10.1109/ACCESS.2023.3325035.